

# Preprocessing

- Unstructured text data
- Text processing is a method to clean the text and prepare it for the model building

# Steps of Preprocessing

- Tokenization
- Stop Word Removal
- Stemming
- Lemmatization

# Tokenization

We break down the text into tokens. Check the example below to see how this is done.

## **Example:**

**Text:** The cat sat on the bed.

**Tokens:** The, cat, sat, on, the, bed

# Issues in Tokenization

- Finland's capital →
  - Finland AND s? Finlands? Finland's?
- Hewlett-Packard → Hewlett and Packard as two tokens?
  - state-of-the-art: break up hyphenated sequence.
  - Co-education
  - lowercase, lower-case, lower case ?
  - It can be effective to get the user to put in possible hyphens
  - San Francisco: one token or two?

How do you decide it is one token?

# Stop Word Removal

- Stop words are used to eliminate words that bear no content or relevant semantics.
- Generally, a stop word list includes articles, pronouns, adjectives, adverbs and prepositions.

## Example:

Text Mining deals with unstructured textual information and it discovers previously unknown structure and implicit meanings buried within the large amount of text. A huge amount of information is present as unstructured text, so we need a special process to analyze it.

## Feature Vector

text 3, unstructured 2, information 2, amount 2,  
and 2, it 2, of 2, a 2,  
mining 1, deal 1, with 1, textual 1, discovers 1, previously  
1, unknown 1, structure 1, implicit 1, meanings 1, buried 1  
within 1, the 1, large 1, huge 1, is 1, present 1, as 1, so 1,  
we 1, need 1, special 1, process 1, to 1, analyze 1

# Issues in Stop Word Removal

Ex: “ This movie is not good.”

If we remove (not ) in pre-processing step the sentence (this movie is good) indicates that it is positive which is wrongly interpreted.

EX: “He lives in US.”

If we remove (in US ) in pre-processing step the sentence (He lives) indicates that it is incomplete sentence.

# Stemming

- Stripping the word to its basic form. The stem can be different from word's linguistic root.
- Advantage: 'read', 'reading' and 'reads' all have same stem 'read'.

## Porter's algorithm

Commonest algorithm for stemming English Results suggest it's at least as good as other stemming options

### Typical rules in Porter

sses → ss

ies → i

ational → ate

tional → tion

# Stemming Algorithm for English

| <b>Word</b>   | <b>Lovin's Stem</b> | <b>Potter's Stem</b> |
|---------------|---------------------|----------------------|
| happier       | hap                 | happier              |
| effectiveness | effect              | effect               |
| happy         | hap                 | happi                |
| Genetic       | genet               | genet                |
| Easy          | ea                  | Easi                 |
| Invisible     | inv                 | Invis                |
| printed       | print               | print                |



# Stemming

## **Example:**

for example compressed and compression are both accepted as equivalent to compress.

## **After Stemming**

for exampl compress and compress ar both accept as equal to compress

# Issues in Stemming

In above example ,

The word “example” is replaced with “exempl “ which is an unknown word and does not have an accurate meaning.

Two different words can have same stem (Example: ‘international’ and ‘internal’ may have same stem ‘intern’.)

# Lemmatization

- Reduce inflectional/variant forms to base form
- the boy's cars are different colors → the boy car be different color
- Lemmatization implies doing “proper” reduction to dictionary headword form
- E.g.,
  - am, are, is → be
  - car, cars, car's, cars' → car
  - Studying -> study